Edugenie- Project Development Phases

6. Project Testing Phase

Testing is performed to verify that all EduGenie features work correctly.

Testing Objectives

- Verify individual modules.
- Check API endpoints.
- Test frontend-backend communication.
- Identify errors.
- Verify AI responses.
- Check invalid input handling.
- Test different types of questions and content.

Functional Testing

Test	Expected Result
Ask a question	Relevant answer displayed
Enter a topic	Simple explanation displayed
Generate quiz	Three MCQs generated
Enter long paragraph	Concise summary generated
Enter learning topic	Learning path generated
Invalid input	Appropriate error message
API failure	Error handled properly

Module Testing

Each module was tested independently before complete integration.

Q&A Testing

Different academic questions were entered to verify the quality and relevance of responses.

Explanation Testing

Complex topics were entered to determine whether the generated explanations were simple and understandable.

Quiz Testing

Quiz generation was tested to ensure that questions contained four options and a valid answer.

Summary Testing

Long paragraphs were used to verify whether important information was retained.

Learning Path Testing

Different subjects were entered to verify beginner-to-advanced recommendations.

Integration Testing

After individual module testing, all modules were connected through FastAPI and tested together.

The frontend was also tested with the backend to ensure that user requests and generated responses were transferred correctly.